

ASSIGNMENT 1

PART C – CLASSES AND RESPONSIBILITIES

Course: B.Tech. CSE

Semester: 5

Subject: System Design (TCS-504)

System: Movie Ticket Booking System

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C. CLASSES AND RESPONSIBILITIES

The following classes are used in the **Movie Ticket Booking System**. Each class has one main responsibility. The data it knows, the work it does, and what it must not do are described below.

1. Classes, Responsibilities and Constraints

Class	What it Knows	What it Does	What it Must NOT Do
Movie	Title, language, duration	Stores and provides movie details.	Must NOT manage seats, bookings or payments.
Seat	Seat number, seat type, price	Represents one physical seat and its price.	Must NOT manage show-specific availability or booking status.
Screen	Screen number, seats	Represents one auditorium and manages its physical seats.	Must NOT handle bookings or payments.
Cinema	Cinema name, screens	Represents the theatre and manages its screens.	Must NOT calculate booking prices or process payments.
Show	Movie, screen, start time, show seats	Represents one screening of a movie and provides its seat layout.	Must NOT process payments or print tickets.

ShowSeat	Seat, status	Maintains the availability/status of a particular seat for a particular show.	Must NOT calculate the total booking price.
Customer	Name, phone	Stores and provides customer information.	Must NOT directly manage payment or seat availability.
Booking	Booking ID, customer, show, seats, total amount, status	Stores and manages one booking and its confirmation/cancellation status.	Must NOT calculate prices or print tickets.
Payment (Abstract)	Payment contract	Defines the common <code>pay(amount)</code> interface for different payment methods.	Must NOT contain implementation of a specific payment method.
UpiPayment	UPI payment details	Processes payment through UPI.	Must NOT manage bookings or ticket printing.
CardPayment	Card payment details	Processes payment through card.	Must NOT manage bookings or ticket printing.
CashPayment	Cash payment details	Processes payment through cash.	Must NOT manage bookings or ticket printing.
PriceCalculator	Seat prices	Calculates the total amount for the selected seats.	Must NOT book seats or process payments.
TicketPrinter	Ticket formatting information	Formats and prints the ticket after successful booking.	Must NOT change booking or payment status.
BookingService	Booking flow and required services	Coordinates seat selection, availability checking, price calculation, payment and booking confirmation.	Must NOT directly implement a specific payment method.

2. Main Methods of Each Class

Movie

getTitle()
getLanguage()
getDuration()

Responsibility: Provides information about the movie selected by the customer.

Seat

getNumber()
getType()
getPrice()

Responsibility: Represents a physical seat and provides its basic details.

Screen

addSeat()
getSeats()
getScreenNumber()

Responsibility: Maintains the physical seats available in a particular screen.

Cinema

addScreen()
getScreens()
getName()

Responsibility: Manages the screens available in the cinema.

Show

findSeat()
showSeatsLayout()
getMovie()
getScreen()
getStartTime()

Responsibility: Represents a particular movie screening and provides its seat layout.

ShowSeat

isAvailable()
book()
release()
getSeat()

Responsibility: Manages the availability of a particular seat for a particular show.

- `isAvailable()` → checks whether the seat is available.
- `book()` → marks the seat as booked.
- `release()` → releases the seat if payment/booking fails.
- `getSeat()` → returns the associated physical seat.

Customer

getName()
getPhone()

Responsibility: Stores and provides customer information.

Booking

addSeat()
setTotalAmount()
confirm()
cancel()
getBookingId()

Responsibility: Represents and manages a customer's booking.

- `addSeat()` → adds a selected seat to the booking.
 - `setTotalAmount()` → stores the calculated booking amount.
 - `confirm()` → confirms the booking after successful payment.
 - `cancel()` → cancels the booking.
 - `getBookingId()` → provides the booking identifier.
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Payment <>

pay(amount)

Responsibility: Defines the common payment interface.

The class is abstract because it does not implement a particular payment method. Specific payment classes provide their own implementation.

UpiPayment

pay(amount)

Responsibility: Processes payment using UPI.

CardPayment

pay(amount)

Responsibility: Processes payment using a card.

CashPayment

pay(amount)

Responsibility: Processes payment using cash.

PriceCalculator

calculateTotal()

Responsibility: Calculates the total amount based on the selected seats and their prices.

TicketPrinter

printTicket()

Responsibility: Generates/formats the ticket details and prints the ticket after booking confirmation.

BookingService

bookTicket()
cancelBooking()

Responsibility: Acts as the main coordinator of the booking process.

bookTicket() coordinates:

Movie Selection



Show Selection



Seat Selection



Availability Check



Price Calculation



Payment



Booking Confirmation



Ticket Printing

cancelBooking() handles cancellation of an existing booking and coordinates the required seat release.

3. Responsibility Distribution

The overall responsibility is divided among the classes to maintain **high cohesion and low coupling**.

Responsibility	Responsible Class
Store movie information	Movie
Represent physical seats	Seat
Manage screen seats	Screen
Manage cinema screens	Cinema
Represent a movie show	Show
Manage show-specific seat status	ShowSeat
Store customer details	Customer
Manage booking information	Booking
Define payment interface	Payment
Process UPI payment	UpiPayment
Process card payment	CardPayment
Process cash payment	CashPayment
Calculate booking amount	PriceCalculator
Print ticket	TicketPrinter
Coordinate complete booking process	BookingService

4. Key Responsibility Rules

The following rules are maintained in the design:

1. **Movie** only manages movie-related information.
 2. **Seat** represents a physical seat and does not know whether it is booked for a particular show.
 3. **ShowSeat** handles show-specific seat availability and booking status.
 4. **PriceCalculator** is responsible only for calculating prices.
 5. **Payment** provides a common abstraction for different payment methods.
 6. **UpiPayment**, **CardPayment** and **CashPayment** handle their respective payment processing.
 7. **Booking** stores booking information and manages confirmation/cancellation.
 8. **TicketPrinter** only prints the ticket and does not modify payment or booking status.
 9. **BookingService** coordinates the complete booking workflow but does not implement a specific payment method.
 10. **Cinema**, **Screen** and **Show** handle the theatre, screen and show structure respectively.
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5. Conclusion

The **Classes and Responsibilities** analysis divides the Movie Ticket Booking System into independent classes, where each class has a clearly defined responsibility.

BookingService acts as the main coordinator, while **PriceCalculator**, **Payment**, **Booking**, **ShowSeat**, and **TicketPrinter** handle their respective specialized tasks. The abstract **Payment** class allows different payment methods such as UPI, card and cash to be used without changing the main booking process. This separation of responsibilities makes the system easier to understand, maintain and extend.